Study notes: Abstract Algebra I, Module 4

Fraleigh, *A First Course in Abstract Algebra* (8th ed.), Module 4 (sections 16, 22 and 23). The syllabus names the 7th edition; items are matched by content. Numbers are the book's own. Student notes in my own words. Every definition, theorem, example, table and figure is kept with its book number. Exercises: the book's exercise sets are not in these notes; the exercises the text names (Section 16 Exercise 10; Section 22 Exercises 30, 36, 37, 38, 45, 50; Section 23 Exercises 26, 35) are printed directly after the item that names them. Figures are described with their labels; not redrawn. Factor groups (Section 12) and factor rings (Section 30) are not taught here; the isomorphism theorems use only the recall stated in Section 16. Sections 17 onward are not in this module.

Section 16: Isomorphism Theorems (16.1–16.10)

Printed pp. 145–148 (16.1–16.10; Figures 16.1, 16.9; Exercise 10 is named in 16.7 and printed on page 149)

Before you start

- Normal subgroups and factor groups G/N (Section 12 of the book, Module V material). Here we only need: if N is normal, G/N is a group whose elements are the cosets gN, and $\gamma : G \to G/N$ with $\gamma(g) = gN$ is a homomorphism.
- Homomorphisms and kernels (Module 2, Section 8; kernel cosets in Module 3, Theorem 10.17).
- Subgroups HN and joins (introduced in this section).

1. The first isomorphism theorem

The isomorphism theorems say that certain factor groups are isomorphic to groups we already know. The first is the most important. Figure 16.1 shows the idea: a homomorphism ϕ from G to G' factors through the factor group G/K, where K is the kernel.

Figure 16.1. A triangle: ϕ runs from G to $\phi[G]$ across the top; γ_K runs from G down to G/K; the dashed arrow μ, labelled "isomorphism", runs from G/K up to $\phi[G]$. Labels as printed; not redrawn.

THEOREM 16.2 (FIRST ISOMORPHISM THEOREM) Let $\phi : G \to G'$ be a homomorphism with kernel K, and let $\gamma_K : G \to G/K$ be the canonical homomorphism. There is a unique isomorphism $\mu : G/K \to \phi[G]$ such that $\phi(x) = \mu(\gamma_K(x))$ for each $x \in G$.

2. Joins and a lemma on normal subgroups

For subgroups H and N of G, let $HN = \{hn \mid h \in H, n \in N\}$. In general HN need not be a subgroup. The **join** $H \vee N$ is the smallest subgroup of G that contains HN (equivalently, both H and N).

LEMMA 16.3 Let N be a normal subgroup of a group G and $\gamma : G \to G/N$ the canonical homomorphism. Then $\phi(L) = \gamma[L]$ is a one-to-one and onto map from the normal subgroups of G containing N to the normal subgroups of G/N.

Idea of the proof. If L is normal and contains N, then $\gamma^{-1}[\gamma[L]] = L$ because L contains every coset xN with $x \in L$. Conversely, if H is normal in G/N, then $\gamma^{-1}[H]$ is normal in G and contains N. ◆

LEMMA 16.4 If N is normal in G and H is any subgroup, then $H \vee N = HN = NH$. If H is also normal, then HN is normal in G.

Proof. Let $h_1, h_2 \in H$ and $n_1, n_2 \in N$. Normality of N gives $n_1 h_2 = h_2 n_3$ for some $n_3 \in N$. Then $(h_1 n_1)(h_2 n_2) = (h_1 h_2)(n_3 n_2) \in HN$. Also $(hn)^{-1} = n^{-1}h^{-1} = h^{-1}n_4 \in HN$. So HN is a subgroup, and by the same argument NH is one; hence $NH = H \vee N = HN$. If H is normal, $ghng^{-1} = (ghg^{-1})(gng^{-1}) \in HN$. ◆

3. The second isomorphism theorem

THEOREM 16.5 (SECOND ISOMORPHISM THEOREM) Let H be a subgroup of G and N a normal subgroup of G. Then $(HN)/N \simeq H/(H \cap N)$.

Proof. N is normal in HN. Define $\phi : H \to HN/N$ by $\phi(h) = hN$. It is a homomorphism, it is onto (every element of HN/N is $hnN = hN$), and its kernel is $\{h \in H \mid hN = N\} = H \cap N$. Apply Theorem 16.2. ◆

Example 16.6. Let $G = \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$, $H = \mathbb{Z} \times \mathbb{Z} \times \{0\}$, and $N = \{0\} \times \mathbb{Z} \times \mathbb{Z}$. Then $HN = \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ and $H \cap N = \{0\} \times \mathbb{Z} \times \{0\}$. So $(HN)/N \simeq \mathbb{Z}$ and $H/(H \cap N) \simeq \mathbb{Z}$.

Example 16.7. Let $G = \mathbb{Z}$, $N = \langle n \rangle$ and $H = \langle h \rangle$ with n, h positive. Write $N + H$ for NH. Then $N + H = \langle \gcd(n, h) \rangle$, since $\gcd(n, h) = xn + yh$ for

some integers x, y. Also $N \cap H = \langle \operatorname{lcm}(n,h) \rangle$. The second isomorphism theorem gives $\langle \gcd(n,h) \rangle / \langle n \rangle \simeq \langle h \rangle / \langle \operatorname{lcm}(n,h) \rangle$. Since a factor group of a cyclic group is cyclic, both sides are isomorphic to $\mathbb{Z}_{n/\gcd(n,h)}$. Comparing orders gives $nh = \gcd(n,h)\operatorname{lcm}(n,h)$.

Exercise 10 (named in Example 16.7). Use one of the Isomorphism Theorems to prove that if $a, b \in \mathbb{Z}^+$ and a divides b, then $|a\mathbb{Z}/b\mathbb{Z}| = b/a$.

4. The third isomorphism theorem

If H and K are normal subgroups of G with $K \leq H$, then H/K is a normal subgroup of G/K.

> **THEOREM 16.8 (THIRD ISOMORPHISM THEOREM)** Let H and K be normal subgroups of G with $K \leq H$. Then $G/H \simeq (G/K)/(H/K)$.
>
> *Proof.* Define $\phi : G/K \to G/H$ by $\phi(gK) = gH$. It is well defined because $gK \subseteq gH$. It is a homomorphism, it is onto, and its kernel is H/K. Apply Theorem 16.2: $\mu((gK)(H/K)) = gH$ is the isomorphism. ◆

Collapsing K and then the H/K cosets gives the same group as collapsing H directly. Figure 16.9 (printed on page 148) shows this as an ellipse for G with the cosets of H drawn as thick-line regions and the cosets of K inside them. The figure carries no labels; it is not redrawn.

Example 16.10. Let $G = \mathbb{Z}_8$, $H = \langle 2 \rangle = \{0,2,4,6\}$, and $K = \langle 4 \rangle = \{0,4\}$. Then

$$G/K = \{\{0,4\},\{1,5\},\{2,6\},\{3,7\}\},$$

$$G/H = \{\{0,2,4,6\},\{1,3,5,7\}\},$$

$$H/K = \{\{0,4\},\{2,6\}\}.$$

Each element of $(G/K)/(H/K)$ is a coset of H/K, so it is a set of sets:

$$(G/K)/(H/K) = \{\{\{0,4\},\{2,6\}\},\{\{1,5\},\{3,7\}\}\}.$$

The isomorphism of Theorem 16.8 sends $\{\{0,4\},\{2,6\}\}$ to $\{0,4,2,6\}$ and $\{\{1,5\},\{3,7\}\}$ to $\{1,5,3,7\}$.

Notes on this section

This section covers book items 16.1 to 16.10 and Figures 16.1 and 16.9. The factor groups it uses are Module V material; the one-paragraph recall above is all the notes need. Exercise 10 is printed after the sentence that names it. The other exercises of Section 16 (1 to 9, and 11) are in the Module 4 excerpt.

Section 22: Rings and Fields (22.1–22.18)

Printed pp. 185–191 (22.1–22.18; Exercises 30, 36, 37, 38, 45, 50 named in the text)

Before you start

- Groups and abelian groups (Modules 1 to 3): the additive part of a ring is an abelian group.
- Integers and their arithmetic: the division algorithm and remainders (Module 2, Theorem 6.2).
- Plus Two recall: the distributive law of multiplication over addition, and $a(b+c) = ab + ac$ in school algebra. (Status: Recall, check against the Kerala text.)

1. Rings

A set with two operations is more general than a group. Such a structure is a **ring**. Hilbert introduced the term; Emmy Noether gave the modern axioms in 1921. The historical note on printed page 186 is kept in the excerpt.

> **DEFINITION 22.1** A **ring** $\langle R, +, \cdot \rangle$ is a set R with two binary operations, addition and multiplication, such that $\langle R, + \rangle$ is an abelian group, multiplication is associative, and for all $a, b, c \in R$ both $a(b+c) = ab + ac$ and $(a+b)c = ac + bc$ hold.

Example 22.2. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ are rings with the usual operations. We write $\mathbb{Z}$ for $\langle \mathbb{Z}, +, \cdot \rangle$ and so on.

Example 22.3. If R is a ring, the $n \times n$ matrices with entries in R form a ring $M_n(R)$. Multiplication is not commutative for $n \geq 2$.

Example 22.4. The set F of all functions $\mathbb{R} \to \mathbb{R}$ is a ring under $(f+g)(x) = f(x) + g(x)$ and $(fg)(x) = f(x)g(x)$. Exercise 36 asks you to check the ring axioms $\mathcal{R}_2$ and $\mathcal{R}_3$ for this ring.

Exercise 36 (named in Example 22.4). Show that the multiplication defined on the set F of functions in Example 22.4 satisfies axioms $\mathcal{R}_2$ and $\mathcal{R}_3$ for a ring.

Example 22.5. $n\mathbb{Z}$ with the usual addition and multiplication is a ring.

Example 22.6. $\mathbb{Z}_n$ with multiplication modulo n is a ring. For example, $(3)(7) = 1$ in $\mathbb{Z}_{10}$.

Example 22.7. The direct product $R_1 \times \cdots \times R_n$ of rings, with operations by components, is a ring.

2. Elementary properties

For a ring R, 0 is the additive identity, $-a$ the additive inverse, and $n \cdot a$ means $a + \cdots + a$ with n summands, with $(-n) \cdot a = (-a) + \cdots + (-a)$. The symbol $n \cdot a$ is not a product of n and a in the ring, since n may not belong to R.

> **THEOREM 22.8** If R is a ring with additive identity 0, then for all $a, b \in R$:
>
> 1. $0a = a0 = 0$;
> 2. $a(-b) = (-a)b = -(ab)$;
> 3. $(-a)(-b) = ab$.
>
> *Proof.* From $a0 + a0 = a(0+0) = a0 = 0 + a0$ the cancellation law gives $a0 = 0$; likewise $0a = 0$. For (2), $a(-b) + ab = a(-b+b) = a0 = 0$, so $a(-b) = -(ab)$; similarly $(-a)b + ab = 0$. For (3), $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$ by (2) and the uniqueness of inverses. ◆

The exercise named in the text after this proof (Exercise 30) asks you to find the error in the proof of (2) that starts from $(-a)b = ((-1)a)b$.

Exercise 30 (named after Theorem 22.8). What is wrong with the following attempt at a proof of Property 2 in Theorem 22.8?
$(-a)b = ((-1)a)b = (-1)(ab) = -(ab)$.

3. Homomorphisms and isomorphisms of rings

> **DEFINITION 22.9** For rings R and R', a map $\phi : R \to R'$ is a **homomorphism** if for all $a, b \in R$: (1) $\phi(a+b) = \phi(a) + \phi(b)$, and (2) $\phi(ab) = \phi(a)\phi(b)$. The **kernel** is $\operatorname{Ker}(\phi) = \{a \in R \mid \phi(a) = 0'\}$.

Condition (1) says ϕ is a group homomorphism of the additive groups, so all results on group homomorphisms apply to the additive part. In particular ϕ is one-to-one exactly when $\operatorname{Ker}(\phi) = \{0\}$. Factor rings are discussed in Section 30 of the book; Module V and later.

Example 22.10. For $a \in \mathbb{R}$ the evaluation map $\phi_a : F \to \mathbb{R}$, $\phi_a(f) = f(a)$, is a ring homomorphism. Finding a real root of $p(x) = 0$ amounts to finding a with $\phi_a(p) = 0$.

Exercise 37 (named in Example 22.10). Show that the evaluation map ϕ_a of Example 22.10 is a ring homomorphism.

Example 22.11. The map $\phi : \mathbb{Z} \to \mathbb{Z}_n$ that sends a to its remainder mod n is a ring homomorphism. The additive part is Module 2 material. For the multiplicative part, write $a = q_1 n + r_1$ and $b = q_2 n + r_2$; then $ab = n(\cdots) + r_1 r_2$, so $\phi(ab)$ is the remainder of $r_1 r_2$, which is $\phi(a)\phi(b)$.

> **DEFINITION 22.12** An **isomorphism** $\phi : R \to R'$ is a ring homomorphism that is one-to-one and onto. The rings are then **isomorphic**.

Isomorphism of rings is an equivalence relation: the inverse of an isomorphism and the composite of two isomorphisms are again isomorphisms. Exercise 38 asks you to check this.

Exercise 38 (named after Definition 22.12). Complete the argument outlined after Definitions 22.12 to show that isomorphism gives an equivalence relation on a collection of rings.

Example 22.13. $\langle \mathbb{Z}, + \rangle$ and $\langle 2\mathbb{Z}, + \rangle$ are isomorphic as groups under $x \mapsto 2x$. This map is not a ring isomorphism, since $\phi(xy) = 2xy$ but $\phi(x)\phi(y) = 4xy$.

4. Units, fields and division rings

$2\mathbb{Z}$ has no identity for multiplication, and matrix rings are not commutative. The ring $\{0\}$ is the **zero ring**. Theorem 22.8 shows that 0 can be a multiplicative identity only in the zero ring.

> **DEFINITION 22.14** A ring is a **commutative ring** if multiplication is commutative. A ring with a multiplicative identity is a **ring with unity**, and the identity is **unity**, written 1.

In a ring with unity, the distributive laws give $(n \cdot 1)(m \cdot 1) = (nm) \cdot 1$.

Example 22.15. If $\gcd(r, s) = 1$, then $\mathbb{Z}_{rs} \simeq \mathbb{Z}_r \times \mathbb{Z}_s$ as rings. The map $n \cdot 1 \mapsto n \cdot (1, 1)$ is an additive isomorphism, and it respects multiplication because $\phi(nm) = (nm) \cdot (1, 1) = [n \cdot (1, 1)][m \cdot (1, 1)]$.

Exercise 45 (named in Example 22.15). Show that the multiplicative inverse of a unit in a ring with unity is unique.

A **multiplicative inverse** of a in a ring with unity $1 \neq 0$ is an element a^{-1} with $aa^{-1} = a^{-1}a = 1$. It is unique when it exists.

> **DEFINITION 22.16** Let R be a ring with unity $1 \neq 0$. $u \in R$ is a **unit** if it has a multiplicative inverse in R. If every nonzero element is a unit, R is a **division ring** (or **skew field**). A **field** is a commutative division ring. A noncommutative division ring is a **strictly skew field**.

Example 22.17. The units of $\mathbb{Z}_{14}$ are $1, 3, 5, 9, 11, 13$. Section 24 shows that the units of $\mathbb{Z}_n$ are the m with $\gcd(m, n) = 1$.

Example 22.18. $\mathbb{Z}$ is not a field: 2 has no inverse. $\mathbb{Q}$ and $\mathbb{R}$ are fields. Section 32 gives a strictly skew field.

A **subring** is a subset that is a ring under the induced operations; a **subfield** is defined the same way for a field. The notation $K \leq L$ means K is a substructure of L.

Exercise 50 (named after Example 22.18). Show that a subset S of a ring R gives a subring of R if and only if $0 \in S$, $a - b \in S$ for all $a, b \in S$, and $ab \in S$ for all $a, b \in S$.

Notes on this section

Book items 22.1 to 22.18 and the named exercises (Exercises 30, 36, 37, 38, 45, 50) are covered. Exercises 1 to 58 of Section 22 are in the Module 4 excerpt. Factor rings and the field of quotients are later material.

Section 23: Integral Domains (23.1–23.14)

Printed pp. 194–196 (23.1–23.14; Figure 23.10; Exercises 26 and 35 named in the text)

Before you start

- Rings, units and fields (Section 22 of the book; notes above).
- Greatest common divisor and relative primeness (Module 2, Section 6).
- Plus Two recall: factorising a quadratic and solving $(x-a)(x-b)=0$. (Status: Recall, check against the Kerala text.)

1. Divisors of zero and cancellation

In the integers a product is 0 only when a factor is 0. We use this when we solve $x^2-5x+6=0$ by factoring it as $(x-2)(x-3)$. This property can fail in other rings.

Example 23.1. In $\mathbb{Z}_{12}$ the factorisation $x^2-5x+6=(x-2)(x-3)$ still holds, but many products are 0: for instance $(2)(6)=0$ and $(3)(4)=0$. So the equation has the solutions $2, 3, 6, 11$ in $\mathbb{Z}_{12}$, not just 2 and 3, since $(6-2)(6-3)=0$ and $(11-2)(11-3)=0$.

DEFINITION 23.2 Two nonzero elements a, b of a ring R with $ab=0$ are **divisors of 0** (or **0 divisors**).

In $\mathbb{Z}_{12}$ the elements $2, 3, 4, 6, 8, 9, 10$ are divisors of 0. These are exactly the numbers whose gcd with 12 is not 1. A unit cannot be a divisor of 0: if $ab=0$ and a is a unit, then $b=a^{-1}ab=0$.

THEOREM 23.3 Let $m \in \mathbb{Z}_n$. Either $m=0$; or $\gcd(m,n)=1$, and m is a unit of $\mathbb{Z}_n$; or $\gcd(m,n)\neq 1$, and m is a 0 divisor of $\mathbb{Z}_n$.

Proof. If $d=\gcd(m,n)\neq 1$ and $m\neq 0$, then $m(n/d)=(m/d)n$ is a multiple of n, so $m\cdot(n/d)=0$ in $\mathbb{Z}_n$ with $m, n/d \neq 0$. If $\gcd(m,n)=1$, write $an+bm=1$ and reduce b modulo n: $b=nq+r$ with $0\leq r\leq n-1$. Then $rm=1-n(a+qm)$, so $rm=1$ in $\mathbb{Z}_n$. ◆

Example 23.4. In $\mathbb{Z}_{20}$ the units are $1, 3, 7, 9, 11, 13, 17, 19$ (since $1\cdot 1=3\cdot 7=9\cdot 9=\cdots=1$), and the 0 divisors are $2, 4, 5, 6, 8, 10, 12, 14, 15, 16, 18$ (for example $2\cdot 10=0$).

COROLLARY 23.5 If p is prime, every nonzero element of $\mathbb{Z}_p$ is a unit; so $\mathbb{Z}_p$ is a field with no divisors of 0.

Proof. For $0<m\leq p-1$, $\gcd(m,p)=1$; apply Theorem 23.3. ■

The book adds that linear algebra over a finite field such as $\mathbb{Z}_p$ uses only field properties, since there is no natural idea of magnitude in $\mathbb{Z}_p$ ($p\cdot 1=0$).

THEOREM 23.6 The cancellation laws hold in a ring R if and only if R has no divisors of 0.

Proof. If cancellation holds and $ab=0$ with $a\neq 0$, then $ab=a0$ gives $b=0$. Conversely, if R has no divisors of 0 and $ab=ac$ with $a\neq 0$, then $a(b-c)=0$ gives $b=c$; the case $ba=ca$ is similar. ◆

In a ring with no divisors of 0, an equation $ax=b$ with $a\neq 0$ has at most one solution. If a is a unit the solution is $a^{-1}b$. In a commutative ring the quotient b/a denotes $a^{-1}b=ba^{-1}$; this notation must not be used in a noncommutative ring.

2. Integral domains

DEFINITION 23.7 An **integral domain** D is a commutative ring with unity $1\neq 0$ that has no divisors of 0.

An integral domain lies between a commutative ring with unity and a field (Example 23.9 and Figure 23.10).

Example 23.8. $\mathbb{Z}$ and $\mathbb{Z}_p$ (p prime) are integral domains. $\mathbb{Z}_n$ is not an integral domain if n is not prime, and a direct product $R\times S$ of nonzero rings is never one, since $(r,0)(0,s)=(0,0)$.

Example 23.9. $\mathbb{Z}_2$ is an integral domain, but $M_2(\mathbb{Z}_2)$ has divisors of 0: $\begin{pmatrix}1&0\\0&0\end{pmatrix}\begin{pmatrix}0&0\\1&0\end{pmatrix}=0$.

Every field is an integral domain, since units are not divisors of 0 and multiplication in a field is commutative.

Figure 23.10. A Venn-style diagram of containment: commutative rings, rings with unity, integral domains, fields, and the regions numbered 2 to 6. Labels as printed; not redrawn.

Exercise 26 (named after Example 23.9, in the sentence that introduces Figure 23.10). Redraw Fig. 23.10 to include a subset corresponding to strictly skew fields.

THEOREM 23.11 Every finite integral domain is a field.

Proof. Let R be a finite integral domain and $a \neq 0$. The map $f(x) = ax$ is one-to-one by cancellation (Theorem 23.6). Since R is finite, f is also onto, so $ab = 1$ for some $b \in R$. Thus a is a unit. ■

The finite condition is needed: $\mathbb{Z}$ is an infinite integral domain that is not a field, and multiplication by 2 is one-to-one but not onto.

3. Characteristic

DEFINITION 23.12 If some positive integer n satisfies $n \cdot a = 0$ for all $a \in R$, the least such n is the **characteristic** of R. If none exists, R has **characteristic** 0.

Exercise 35 (named after Definition 23.12). Show that the characteristic of an integral domain D must be either 0 or a prime p. [*Hint:* If the characteristic of D is mn, consider $(m \cdot 1)(n \cdot 1)$ in D.]

Example 23.13. $\mathbb{Z}_n$ has characteristic n; $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ have characteristic 0.

THEOREM 23.14 Let R have unity. If $n \cdot 1 \neq 0$ for all $n \in \mathbb{Z}^+$, then R has characteristic 0; otherwise the least n with $n \cdot 1 = 0$ is the characteristic.

Proof. If $n \cdot 1 = 0$, then $n \cdot a = a(n \cdot 1) = a0 = 0$ for every $a \in R$. ◆

Notes on this section

Book items 23.1 to 23.14 are covered, with Figure 23.10 and the named exercises (26 and 35). Exercises 1 to 36 of Section 23 are in the Module 4 excerpt.